



# UTM

UNIVERSITI TEKNOLOGI MALAYSIA

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**BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)**

**SECV 2113 – HUMAN COMPUTER INTERACTION**

**EV CHARGING STATIONS TRACKER**

**TASK 1**

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## **EV Charging Stations tracker**

Explain for the name we chosen "But U"

The main reason we choose the name "But U" is because it is catchy, easy to remember, and stands out, which makes it memorable for users. It also acts as a reminder that EV driving is not just about transportation; it's a small but impactful choice toward climate action and sustainability. While many people still use petrol-powered cars, "But U" can start the first step to help the environment by reducing emissions, and embracing a cleaner, greener future..

### **Introduction**

As the rapid growth of electric vehicle (EV) adoption represents a critical step toward sustainable transportation and reducing global greenhouse gas emissions. However, as more consumers shift to electric vehicles, the demand for accessible and reliable EV charging infrastructure has significantly increased. Many EV owners still face challenges locating available charging stations, especially on long journeys like highway or in rural areas, which can lead to "range anxiety", a common worry about running out of charge before reaching a station, which takes away from the convenience and freedom of driving an EV.

This project proposes an innovative EV charging station tracker app designed to meet the needs of current and prospective EV drivers. By providing real-time information on charging station locations, availability, and compatibility, the app aims to make EV charging as easy and worry-free as filling up a gas tank. Furthermore, integrated route planning and user reviews are intended to eliminate common pain points associated with EV charging, ensuring that users can confidently travel with access to dependable charging resources.

By supporting the transition to clean energy and aligning with Sustainable Development Goals SDG-7 which is Affordable and Clean Energy and SDG-13 (Climate Action), this app seeks to empower users, reduce environmental impact, and contribute to a future where electric vehicles become a mainstream and sustainable choice for everyone.

## Problem Description

Electric vehicle (EV) drivers face significant challenges with charging infrastructure, especially with the growing adoption of EVs. Common issues include difficulty locating charging stations, unreliable real-time status updates, and lack of route optimization. These problems lead to delays, “range anxiety,” and reduced user confidence in EV use. Furthermore, inconsistent charging costs and payment options add to the inconvenience, potentially deterring EV adoption, which impacts climate action and sustainability goals.



## Problems of the Existing System

### 1. Limited and Inconsistent Charging Infrastructure

**Usability Goal:** Effectiveness and Efficiency

**User Experience Goal:** Safety and Comfort

Many EV charging stations are not evenly distributed, especially in rural areas or along highways. This inconsistency results in long detours or extended waits for charging, impacting user travel plans and increasing stress.

### 2. Outdated or Unavailable Real-Time Information

**Usability Goal:** Learnability and Accessibility

**User Experience Goal:** Reliability

Current systems often fail to provide real-time updates on charger status (occupied, out of service), leaving users guessing. This lack of accurate information undermines user trust and disrupts efficient route planning.

### 3. Inconsistent Charging Costs and Payment Methods

**Usability Goal:** Efficiency

**User Experience Goal:** Satisfaction

The absence of standardized pricing and payment options across different charging providers creates confusion and inconvenience. Users might face unexpected costs or find themselves unable to pay due to limited payment methods.

## Proposed Solutions

First of all, BUT U will develop a map-based feature that shows current availability of nearby charging stations with filters for charger types, charging speed, power levels and compatibility with different EV models. BUT U is integrating data from various charging networks will allow users to see if a station is occupied, in use, or temporarily out of service, minimizing guesswork and maximizing convenience. It will also suggest optimized routes with available charging stops based on vehicle range and energy needs to avoid wasted trips to occupied or malfunctioning stations.

Besides, BUT U will offer standardized information on charging costs and payment options. System will offer clear, consistent information on charging costs and accepted payment methods for each station. It will support multiple payment methods, including mobile wallets, credit/debit cards, and subscription-based access, providing a seamless transaction experience across multiple charging providers.

Furthermore, BUT U provides customizable alerts and notifications for users, such as when a preferred charging station becomes available or when their EV requires charging based on typical travel patterns. This feature could also alert users to any changes in a station's availability or issues reported by other users, adding convenience and reliability to the EV charging experience.

BUT U also incorporates a user review and rating system where users can share feedback on individual charging stations. This feature allows users to assess station reliability, cleanliness, and additional amenities, fostering a community-driven approach to maintaining charging infrastructure standards.

Last but not least, to enhance the user experience, BUT U offers a eco-points system and gamification function to allow users earn eco-points for actions such as charging stations or planning eco-friendly routes. And introduce fun challenges like "Charge at 5 different stations in a week" with rewards for completion. These rewards and points can be redeemed for discounts, eco-friendly products or charging credits.

## **Target Users**

As the demand for EV stations is increasing, This system is aimed for EV owners or drivers, potential EV buyers, fleet managers and delivery drivers, travelers and road trippers with EVs, environmentally conscious individuals and real estate developers and property managers.

### **Electric Vehicle Owners and Drivers**

Individuals who use electric vehicles as their primary means of transportation for daily commuting require frequent charging station access along their routes especially during the peak hours. This application will help them to quickly locate available stations and reserve slots so that it can minimize the disruptions to their routine.

### **Electric Vehicle Companies**

EV manufacturers and providers, such as BYD can use the system to enhance their customer offerings. By integrating this application into their services, these companies can provide users with seamless access to charging stations, promote their brand's charging networks, and collect valuable insights into charging behavior to refine their infrastructure planning and improve customer satisfaction.

### **Travelers and Road Trippers with EVs**

Tourists or individuals driving long distances may be less familiar with charging station locations and need reliable options during their trips. This application will serve as a travel companion to help them to find suitable charging spots, plan routes with reliable station access and also gain insights on amenities available at each location.

## Ideas from ChatGPT

### Gamification & Rewards System

1. **Eco-Points System:** Reward users with eco-points for charging their EVs at stations or planning efficient routes that minimize emissions. Points can be redeemed for discounts, eco-friendly products, or charging credits.
2. **Charging Challenges:** Introduce challenges, such as "Charge at 5 different stations in a week," with rewards for completion.

### Enhanced Features with AI

3. **Predictive Availability:** Use AI to predict station availability based on historical data, traffic patterns, and station usage trends, giving users an estimated wait time for a station.
4. **Personalized Recommendations:** Suggest stations based on the user's past preferences (e.g., nearby coffee shops, rest areas) or their typical charging behavior.

### Social & Community Features

5. **EV Community Hub:** Allow users to connect with other EV drivers to share tips, reviews, and updates on charging station conditions or amenities.
6. **Carpooling Integration:** Provide a feature for EV carpoolers to coordinate stops at charging stations together, enhancing resource use efficiency.

### Environmental Impact Tracking

7. **Carbon Offset Integration:** Let users track how much CO2 they've saved by using their EV, and integrate options to donate to carbon offset programs directly through the app.

### Smart Charging Assistance

8. **Dynamic Charging Alerts:** Notify users of off-peak times when charging is cheaper or greener due to renewable energy availability in the grid.
9. **Scheduled Charging:** Allow users to reserve a charging slot and start charging automatically during off-peak hours.

### Entertainment & Productivity

10. **Charging Station Engagement:** Offer station-specific content such as AR games, trivia, or virtual tours while users wait for their car to charge.
11. **Productivity Tools:** Integrate workspace tools like Wi-Fi availability information, nearby cafes with charging spots, or coworking spaces.

## Customization & Accessibility

12. **Adaptive Route Planning:** Factor in weather conditions, elevation changes, and traffic to optimize energy use during trips.
13. **Universal Charging Adaptor Finder:** Suggest places to buy/rent charging adaptors if a station isn't compatible with the user's vehicle.

## Additional Safety Features

14. **Emergency SOS:** Include a feature to notify roadside assistance or nearby EV users if a driver runs out of charge in a remote area.
15. **Real-Time Security Updates:** Provide safety alerts for stations in low-light or isolated areas based on user reviews.